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instance	a ₁	a ₂	a ₃	Target class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	f	5.0	-
4	f	f	4.0	+
5	f	T	7.0	-
6	f	T	3.0	-
7	T	f	8.0	-
8	f	T	7.0	+
			5.0	-

$$\text{Entropy} = - \sum_{i=1}^n p_i \log_2(p_i)$$

Prob. of i^{th} class

4 are +ve, 5 are -ve

$$= -\frac{4}{9} \log_2(4/9) - \frac{5}{9} \log_2(5/9)$$

$$= 0.9911$$

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$$\text{Info gain} = \text{Entropy} + \text{Entropy}_F, +ve$$

$$\text{Entropy}(T) = -\frac{3}{4} \log_2(3/4)$$

$$-\frac{1}{4} \log_2(1/4) \quad \leftarrow -ve$$

$$= 0.311 + 0.5$$

$$= 0.811$$

$$\text{Entropy}_F = -\frac{1}{5} \log_2(1/5) - \frac{4}{5} \log_2(4/5)$$

$$\text{Info. gain} = 1 - 0.9911 = -\text{Ent}(T)$$

$$- \text{Ent}(F)$$

$$1 - 0.9911 = 0.0089$$

$$= 0.9911 - 0.811$$

$$= 0.1801$$

$$= 0.229$$

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$$\text{Infor. gain} = \text{Entropy}_T + \text{Entropy}_F$$

$$\text{Entropy}(T)$$

$$= -\frac{2}{5} \log(2/5)$$

$$= -\frac{3}{5} \log(3/5)$$

$$\text{Entropy}(F)$$

$$= -\frac{2}{4} \log(2/4) - \frac{2}{4} \log(2/4)$$

$$\text{Infor. gain } a_2 = 0.9911 - \text{Entropy}(T) - \text{Entropy}(F)$$

$$\text{Entropy}(T) = 0.9911$$

$$\text{Entropy}(F)$$

$$= 0.9911 - 0.9709$$

$$1.180 - 1.1990$$

$$= 1.0$$

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Gini of a1

Gini Index of a1 & a2

$$\text{Gini} = 1 - \sum_{i=1}^n (P_i)^2$$

$$\text{Gini(True)} = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2$$

$$\text{Gini(false)} = 1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2$$

$$\text{Gini} = \frac{4}{9} \times \text{Gini(T)} + \frac{5}{9} \times \text{Gini(F)}$$

$$= \frac{4}{9} \times 0.375 + \frac{5}{9} \times 0.32$$

Gini of a2

$$\text{Gini(T)} = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2$$

$$\text{Gini(F)} = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2$$

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Gini index of a_2

$$= \frac{4}{9} \times \text{Gini}(T)$$

$$+ \frac{5}{9} \times \text{Gini}(F)$$

select split attribute between

a_1 and a_2

$$\text{Gain}(a_1) = -0.295$$

$$\text{Gain}(a_2) = -0.0072$$

Higher information gain produces

better split

Hence attribute a_1 is best

Split attributes depend on information gain

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a_1 is the best split attribute

Gini index of $a_1 = 0.344$

Gini index of $a_2 = 0.488$

Small gini index

produces better split

Hence a_1 is the best split

attribute

$$\frac{1}{N} \log \frac{1}{2} - \left(\frac{1}{N} \right) \log \frac{1}{2} - \left(\frac{1}{N} \right) \log \frac{1}{2}$$

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study Result

low fail

low fail

high pass

high pass

high pass

low fail

high pass

low fail

prob of i th class
 n

$$\text{Entropy} = - \sum_{i=1} p_i \log p_i$$

$$= - \frac{4}{8} \log \left(\frac{4}{8} \right) - \frac{4}{8} \log \left(\frac{4}{8} \right)$$

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$$P(\text{pass}) = \frac{4}{8}$$

$$P(\text{Fail}) = 4/8$$

Gini Index

$$\text{Gini} = 1 - \sum p_i^2$$

$$= 1 - (0.5)^2 - (0.5)^2$$

3. Split on study hrs

low : 4 records : All fail

High : 11 : All pass

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Naives Bayes classification

SNO	color	legs	Ht	Smell
1	white	3	Short	yes
2	green	2	tall	no
3	green	3	S	yes
4	white	3	S	yes
5	green	2	S	No
6	white	2	T	No
7	white	2	T	No
8	white	2	S	yes

Estimate conditional probability of each attribute, color, legs, Ht, Smelling using data given below

Prior prob = $\frac{4}{8}$ $P(H) = \frac{4}{8}$

PCM)

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Target Species

M
M
M
M
H
H
H
H

color

white

green

M

$\frac{2}{4}$

$\frac{2}{4}$

H

$\frac{3}{4}$

$\frac{1}{4}$

cond. prob

legs

2

3

M

$\frac{1}{4}$

$\frac{3}{4}$

H

$\frac{4}{4}$

0

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	M	H
Height		
short	$3/4$	$2/4$
tall	$1/4$	$2/4$

	M	H
Smelling		
yes	$3/4$	$1/4$
No	$1/4$	$3/4$

Prob that color = white given $= 2/4$

Prob that color = green given $= 2/4$

$$P(M/\text{new instance}) = P(M) \times P(\text{color} = \text{green} | M) \times (P(\text{legs} = 2/n))$$

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$$\times P(\text{ht} = \text{tall} / H) \times P(\text{smell} = \text{no} / H)$$

$$= 2/4 \times 2/4 \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

$$= 0.5 \times 0.5 \times 0.25 \times 0.25 \times 0.25$$

$$= (0.25)^5 = 0.0009$$

$P(H / \text{new instance})$

$$= P(H) \times P(\text{color} = \text{green} / H)$$

$$\times P(\text{leaves} = 2 / H)$$

$$\times P(\text{ht} = \text{tall} / H)$$

$$\times P(\text{smell} = \text{no} / H)$$

$$= 0.5 \times 0.25 \times 1 \times 0.5$$

$$\times 0.75$$

$$= 0.047$$